Lab 2 - Responsive Navbar and Grid Drill

Completion Report

Name: Sian Ahmed Asif

Roll Number: FA24-BCS-118

Folder: lab-2/

1. Goal of the Lab

The goal of this lab was to practice responsive design before Assignment 2. I had to build a navbar and a product grid that change their layout depending on the screen width. The content is only placeholder text and images because the lab is about how the layout reacts, not about the final design.

2. Requirements

Part	Phone (under 768px)	Tablet (768px+)	Desktop (992px+)
Navbar	Logo and hamburger button	Logo left, links center, cart right	Logo left, links center, cart right
Product grid	1 column	2 columns	4 columns

3. Tools and Files

• **HTML** for the structure of the page (index.html).

• **Bootstrap 5.3.3** for the navbar. I saved its files in the bootstrap folder so the page also works without internet.

• **CSS Grid** for the product grid, written in my own file style.css.

• **Media queries** to change the number of columns at different widths.

• **Images** folder with 8 simple placeholder pictures.

4. How I Built the Navbar

I used the Bootstrap navbar. The class **navbar-expand-md** is the most important part. It tells Bootstrap to show the full menu when the screen is 768px or wider and to hide it behind a hamburger button when the screen is smaller.

The logo is the first item, so it stays on the left. I put the links in a list with the class **mx-auto**. This adds automatic margin on both sides, which pushes the links to the center and pushes the cart to the right. The hamburger button uses **data-bs-toggle="collapse"**, so Bootstrap's JavaScript file is needed for it to open and close.

```
<nav class="navbar navbar-expand-md navbar-dark bg-dark px-3">
  <a class="navbar-brand" href="#">MyShop</a>
  <button class="navbar-toggler" data-bs-toggle="collapse"
          data-bs-target="#menu">...</button>
  <div class="collapse navbar-collapse" id="menu">
    <ul class="navbar-nav mx-auto"> links </ul>
    <a class="nav-link text-white">Cart</a>
  </div>
</nav>
```

5. How I Built the Product Grid

I made a div with the class **product-grid** and used **display: grid**. I wrote the phone version first, with one column, because that is the default. Then I used two media queries to add more columns when the screen gets wider. **repeat(4, 1fr)** means four columns that all have the same width, and **gap** adds space between the cards.

```
.product-grid {
  display: grid;
  grid-template-columns: 1fr;
  gap: 20px;
}

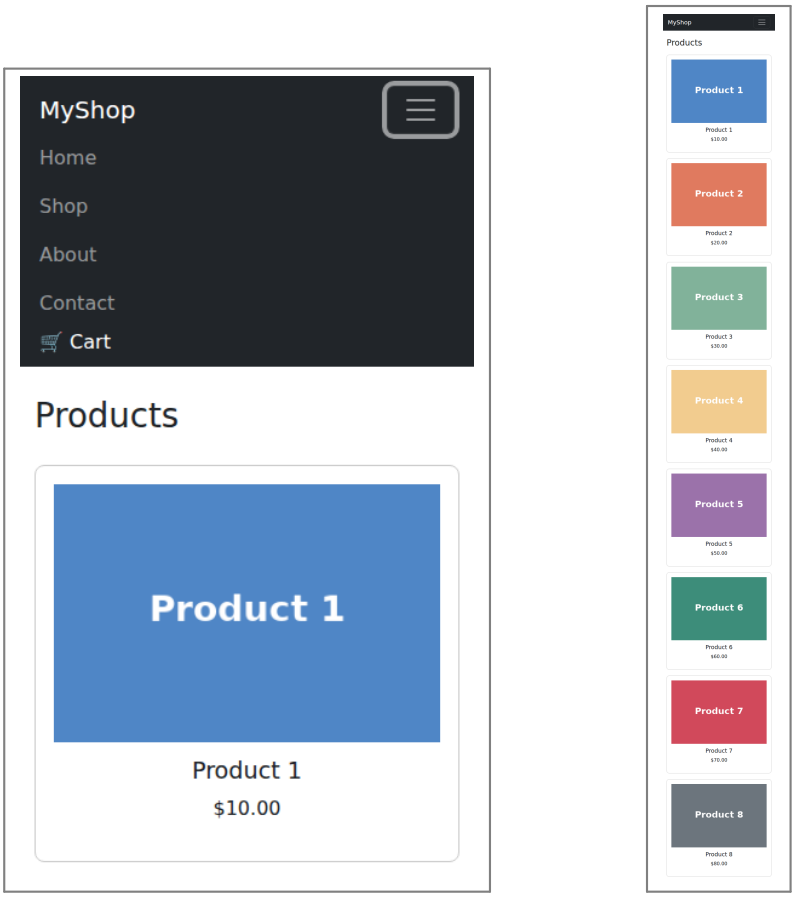
@media (min-width: 768px) {
  .product-grid { grid-template-columns: repeat(2, 1fr); }
}

@media (min-width: 992px) {
  .product-grid { grid-template-columns: repeat(4, 1fr); }
}
```

6. Screenshots

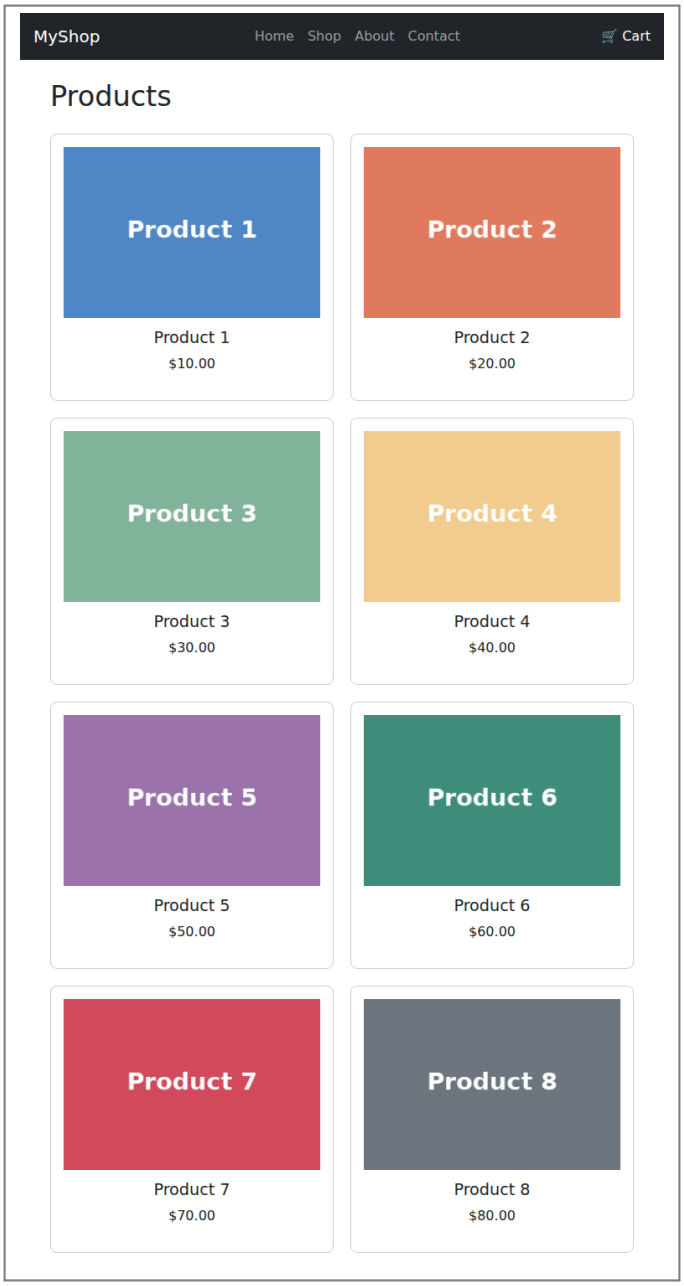
I took the screenshots with the browser developer tools (device mode) at three widths.

6.1 Phone - 375px wide (1 column)



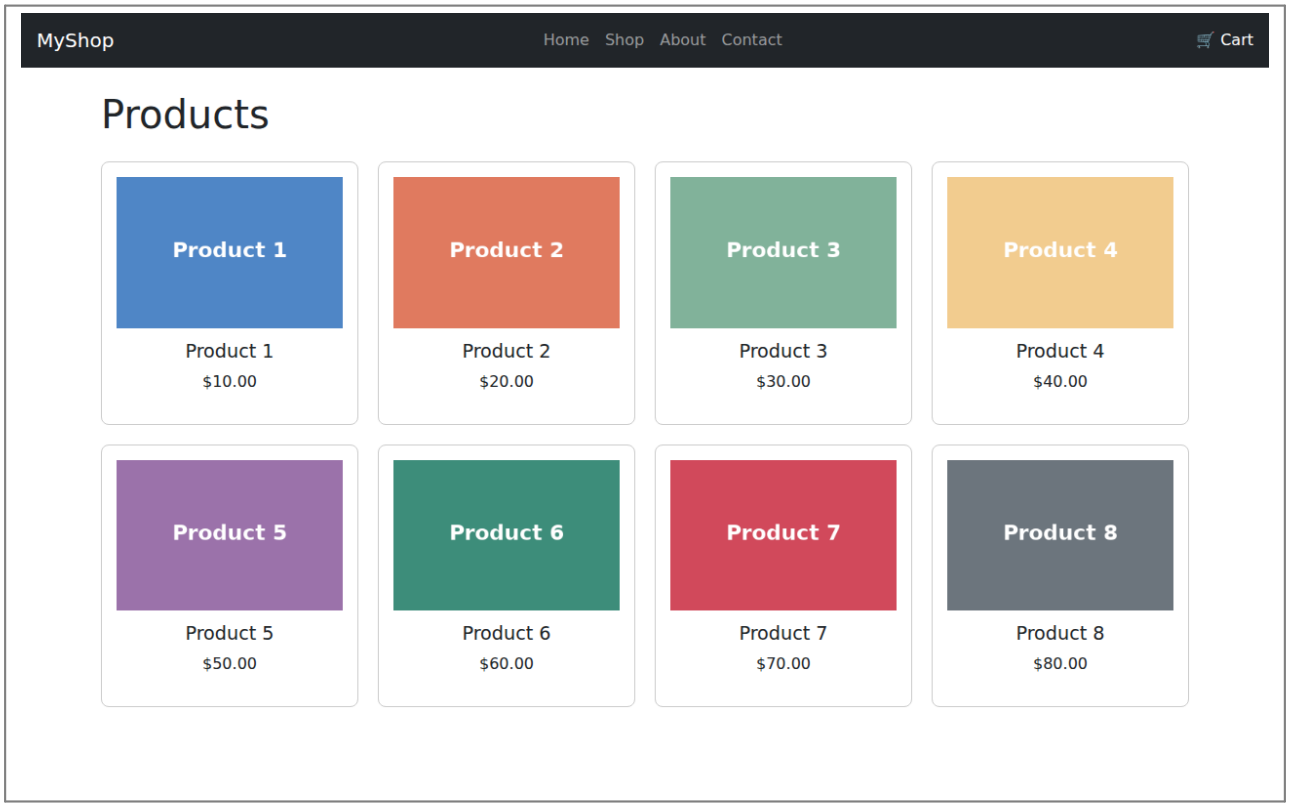
Left: hamburger menu opened. Right: full page with the cards in one column (scaled down to fit).

6.2 Tablet - 768px wide (2 columns)



The full menu is visible and the products are in two columns.

6.3 Desktop - 1280px wide (4 columns)



Logo on the left, links in the center, cart on the right, and four products in each row.

7. Problems and What I Learned

• At first the product names looked too big because the h2 tag has a large default size. I fixed it by setting a smaller font-size in style.css.

• The hamburger button did nothing until the Bootstrap JavaScript file was linked at the bottom of the page.

• The viewport meta tag is required. Without it, phones show the desktop layout zoomed out and the media queries do not work properly.

• Starting with the phone layout and adding media queries for bigger screens made the CSS shorter and easier to understand.

8. Conclusion

In this lab I learned how a navbar can collapse into a hamburger menu and how a grid can change from 1 to 2 to 4 columns using media queries. I will use the same ideas in Assignment 2 with the real content and design.